
Supporting information

AI as an Amplifier in Chemistry Laboratories: A SERG-Based Framework for Embodied and Assessable Instruction

Zhaohong Zuo^{a,*}, Jun Li^a

5 ^a School of Chemistry and Chemical Engineering, Chongqing University, Chongqing 401331, China
(ORCID 0000-0002-2460-8702)

* E-mail: zzhong@cqu.edu.cn

Appendix S1 (SI). Sample Student Work (T0 vs. T2, De-identified)

Purpose. To illustrate how formative assessment and project-based organization address the problem of “missing evaluation,” we compare baseline (T0) and post-intervention (T2) laboratory reports. The contrast showcases qualitative evidence of evidence-chain construction and supports the **Evidence** dimension in the SERG model.

Sample selection. Two de-identified reports from the experimental group (EG) are randomly selected for the HPLC experiment: one at **T0** (before intervention, without rubrics and AI process map) and one at **T2** (after intervention, with embedded formative assessment and a project-based unit). All samples were processed for **FERPA/GDPR** compliance; personally identifiable information was removed.

T0 Sample (Baseline)

Excerpt.

“Measured sample concentration; obtained peak areas: 0.234, 0.245, 0.251. Calculated concentration is 0.05 mol/L. The experiment was successful and the data are reliable.”

Characteristics.

- Loosely structured: lists data and a conclusion, but lacks method description, uncertainty analysis, and reflection.
- No discussion of error sources or propagation; the evidence chain is incomplete.
- Reveals the “missing evaluation” issue: the report stops at results without evidence-based explanation.

T2 Sample (Posttest)

Excerpt.

“**Objective:** Determine sample concentration by HPLC and verify method reliability. **Method:** C18 column, flow rate 1.0 mL/min, detection at 254 nm, three replicates; peak areas 0.234 ± 0.005 , 0.245 ± 0.004 , 0.251 ± 0.006 . **Data processing:** Linear regression ($R^2 = 0.998$); concentration 0.050 ± 0.002 mol/L. **Uncertainty analysis:** Major contributors are injection volume (± 0.1 μ L, 70%) and instrument precision (± 0.005 AU, 25%). **Conclusion:** Result is reliable; 95% CI = 0.048–0.052 mol/L. **Reflection:** Improve injection precision to reduce uncertainty.”

Characteristics.

- Structured presentation (objective—method—data—uncertainty—conclusion—reflection) aligned with the rubric.
- Includes uncertainty sources, quantitative estimates, and propagation; the evidence chain is complete.
- Shows AI-assisted analysis (e.g., fitting and visualization), enabling assessment-driven learning and mitigating the “missing evaluation” problem.

Comparative Analysis

The **T0** sample reflects “misalignment” and “missing evaluation,” focusing on outcomes without integrating technique and disciplinary reasoning. The **T2** sample—organized through high-quality project-based learning and guided by formative rubrics—substantially improves the evidence chain and demonstrates AI as an **amplifier** for data analysis and reflection, supporting the **Evidence** and **Growth** dimensions of SERG.

Appendix S2 (SI). Interview Coding Themes

Purpose. Through thematic coding of semi-structured interviews, this appendix reveals how teachers and students perceived the intervention, supplements quantitative findings, explains mechanisms for addressing the three challenges (misalignment, instability, missing evaluation), and supports the **Reusability** and **Growth** dimensions of the SERG framework.

Method. Semi-structured interviews were conducted with 10 teachers in the experimental group and 30 students (three randomly sampled per class). Questions centered on the three challenges (misalignment, instability, missing evaluation) and intervention experiences (TPACK alignment, dual-channel fallback, project-based evaluation). Using thematic analysis (open → axial → selective coding), we derived the following axial themes and subthemes. Quotations are de-identified.

Coding Themes

1. Axial Theme: Technological Alignment (addressing misalignment)

- **Subtheme 1: Technology–pedagogy integration**

- Illustrative quote (Teacher T3, Analytical Chemistry):**

- “Previously, AI was used haphazardly—more for show than for learning. With the TPACK handbook, lesson planning now makes explicit how technology serves the experimental objective; for example, using AI curve-fitting to guide students to discuss sources of error.”

- Analysis:* Reflects how TPACK × SAMR reduces “technology-for-technology’s-sake” and improves task alignment.

- **Subtheme 2: Progression along SAMR**

- Illustrative quote (Teacher T7, Physical Chemistry):**

- “We moved from simple Excel substitution to AI-driven kinetics visualization. Students no longer just look at numbers; they actively explore the meaning of parameters.”

- Analysis:* Demonstrates movement from “substitution” toward “modification/redefinition,” supporting reusability.

2. Axial Theme: Classroom Stability (addressing instability)

- **Subtheme 1: Dual-channel fallback**

- Illustrative quote (Student S12, EG class):**

- “When the cloud froze, the teacher switched to a local Excel script in 30 seconds. The experiment didn’t stop, which felt reassuring.”

- Analysis:* UDL 3.0 and the dual-channel design lower failure rates and sustain instructional continuity.

- **Subtheme 2: Increased technological confidence**

- Illustrative quote (Teacher T5, Organic Chemistry):**

- “With the fallback scripts, I’m no longer worried about device glitches; my teaching pace is steadier.”

- Analysis:* Enhanced stability builds teacher confidence in technology use, reinforcing the **Stability** dimension of SERG.

3. Axial Theme: Evidence-Chain Construction (addressing missing evaluation)

- **Subtheme 1: Immediate feedback through formative assessment**

- Illustrative quote (Student S25, EG class):**

- “The rubrics showed me I needed to report error sources, and AI’s real-time feedback helped me improve uncertainty analysis.”

- Analysis:* Formative assessment plus AI feedback mitigates “missing evaluation” and elevates data literacy.

- **Subtheme 2: Project-based organization fosters reflection**

- Illustrative quote (Student S8, EG class):**

- “The project made me think through the whole process—from data collection to explanation. After the public presentation, peer review made me more careful about the rigor of my conclusions.”

- Analysis:* HQPBL and rubrics strengthen the evidence chain, supporting the **Evidence** dimension of SERG.

Summary. The interview themes explain how the intervention reduces **misalignment** through TPACK × SAMR alignment, enhances **stability** via dual-channel fallback grounded in UDL 3.0, and remedies **missing evaluation** through project-based assessment with rubrics. These qualitative findings triangulate with the quantitative results in Tables 1–3 and substantiate the intervention’s **Reusability** and **Growth** for broader adoption.

Appendix S3 (SI). Classroom Observation Logs

Purpose. To provide process evidence of the intervention in action and verify how the **alignment–stabilization–evidence–safeguards** chain addresses the three persistent challenges. These logs complement quantitative analyses and support the **Stability** and **Evidence** dimensions of the SERG framework.

95 **Method.** Observers (research team members) conducted structured observations of **20 class sessions** (90 minutes each) across **10 experimental-group classes** at **T1 (Weeks 3–4)** and **T2 (Weeks 5–14)**. Key events were recorded, including technology use, failures and responses, and assessment feedback. Logs were de-identified to comply with privacy requirements. Representative excerpts follow.

Log Samples

- 100 • **Class:** EG-3 (Analytical Chemistry, HPLC experiment, T2 Week 8)
- **Time:** 14:20–14:25
 - **Event:** The cloud-based AI platform stalled due to network fluctuation. The teacher executed the contingency plan and switched to a local Excel script within **30 seconds**, allowing students to continue curve fitting.
 - 105 ○ **Observer note:** “The switch was smooth and did not disrupt pacing. Students showed no evident anxiety and continued discussing the relationship between peak area and concentration.”
 - **Linked challenge: Instability** (SERG: **Stability**) — the dual-channel fallback effectively reduced interruption risk.
- 110 • **Class:** EG-7 (Physical Chemistry, kinetics experiment, T2 Week 10)
- **Time:** 15:10–15:30
 - **Event:** Student teams used AI tools to visualize rate constants and, guided by the rubric for self-assessment, identified error sources (temperature control ± 0.1 °C). Peer review then informed refinement of report conclusions.
 - 115 ○ **Observer note:** “Students actively cited the uncertainty prompt card. Discussion shifted from ‘is the result correct’ to ‘how does error affect the conclusion.’ AI feedback flagged data deviations in real time; the teacher provided targeted prompts.”
 - **Linked challenge: Missing evaluation** (SERG: **Evidence**) — formative assessment plus AI support strengthened the evidence chain.
- 120 • **Class:** EG-1 (Organic Chemistry, titration experiment, T1 Week 4)
- **Time:** 13:45–14:00
 - **Event:** Guided by the TPACK handbook, the teacher delivered a micro-lesson. The AI process map led students from data import to curve fitting; the task sheet specified **objective–method–reflection** requirements.
 - 125 ○ **Observer note:** “Students followed the workflow; lesson design and enactment were highly aligned. AI tools were embedded in the experimental logic. Student questions focused on technical details rather than procedural confusion.”
 - **Linked challenge: Misalignment** (SERG: **Reusability**) — technology–pedagogy alignment reduced “technology for technology’s sake.”

Summary

130 The observation logs substantiate implementation fidelity: **dual-channel switching** (≈ 30 – 45 s) secured classroom stability; the integration of **AI and rubrics** improved the **evidence chain**; and **TPACK-informed alignment** optimized task design. These records triangulate with **Table 2** (declining failure rates), **Table 3** (improved report quality), and **Table 4** (stronger lesson–enactment alignment), demonstrating a system-level resolution of the three challenges and supporting SERG’s closed-loop validation.